

An Efficient Multi-View Multimodal Data Processing Framework for Social Media Popularity Prediction

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ABSTRACT

Popularity of social media is an important symbol of its communication power. Predictions of social media popularity have tremendous business and social value. In this paper, we propose an efficient multimodal data processing framework, which can comprehensively extract the multi-view features from multimodal social media data and achieve accurate popularity prediction. We utilize Transformer and sliding window average to extract time series features of posts, utilize CatBoost to calculate the importance of different features, and integrate important features extracted from multiple views for accurate prediction of social media popularity. We evaluate our proposed approach with the Social Media Prediction Dataset. Experimental results show that our approach achieves excellent performance in the social media popularity prediction task.

CCS CONCEPTS

• **Computing methodologies** → **Information extraction; Computer vision; Scene understanding**; • **Networks** → **Network reliability**.

KEYWORDS

Social Media Popularity Prediction, Transformer, CatBoost, Feature Selection

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1 INTRODUCTION

With the rapid development of the Internet, social media has become an important part of modern work and life. In order to increase the flow of the social media platforms and gain more economic benefits, many platforms push interesting news, emerging topics, latest stories or amazing products to attract internet users to watch them online. At the same time, people have a herd mentality, so that the popularity of social media can reflect the thinking of the general public. That is to say, predicting the popularity of social media is also predicting what people are likely to think in the future, to some extent. If we can know which post has the highest popularity, we can know the trend of the network, so that we can promote corresponding products or do other things to obtain good social or economic benefits. However, due to the huge amount of information on the network, it is impossible to manually screen potential hot posts, so it is a new and important task to automatically identify potential hot posts. Wu *et al.* [25, 26, 27] formulated this task as the social media popularity prediction, which focuses on predicting the impact of online post sharing on social media.

In recent years, many researchers have explored research on predicting the popularity of social media. At the model level, Wu *et al.* proposed Deep Temporal Context Networks (DTCN) [26], which treats posts as a sequential post stream. Liu *et al.* [15] proposed a memory-efficient neural architecture search framework, which can find a suitable model for popularity prediction. At the feature level, Chen *et al.* [3] proposed a framework that uses visual-textual features trained by XGBoost [4] for popularity prediction. Gelli *et al.* [8] used visual sentiment features together with context features to predict the popularity score of social images. Lv *et al.* [16] investigated and extracted visual and social features of images. Wang *et al.* [24] explored the sequential relevance between user content and proposed a method to extract textual features. Lin *et al.* [14] leveraged the self-attention mechanism to fuse different features of the post. Shang *et al.* [20] propose a pre-trained framework for popularity prediction. [12, 22] use CatBoost as their basic model for the prediction of popularity.

Our work is carried out at the level of feature. We believe that it is important to analyze the available social media data from multiple perspectives to better predict its popularity, like thinking as a real person. The above methods only use a single algorithm to extract features (single-view), which cannot take into account the different emphases extracted by different algorithms (multi-view).

Meanwhile, they also lack the extraction of timing features. Therefore, we propose an efficient multi-view multimodal data processing framework for social media prediction, which uses different algorithms to extract features from different views. Then we select and use the most important features for predictions of social media popularity, which can make full use of existing social media information while reducing the dimensionality of features. At the same time, we also propose a time series feature extraction method, which can enhance the distribution consistency of the feature of training set and test set. The main contributions of this paper are as following:

- We propose an efficient multi-view multimodal data processing framework, which can cascade multiple algorithms while preventing memory overflow, improve the adequacy of feature extraction, and improve the performance of the prediction model.
- We propose a Transformer based time series feature extraction method to extract the time series features of user posts and ensure the consistency of feature distribution between the training set and test set, which can improve the time series modeling.
- We evaluate the proposed approach using the Social Media Prediction Dataset from the Social Media Prediction Challenge of the ACM Multimedia Conference. Experimental results show its powerful performance for social media popularity prediction.

2 METHOD

In this section, we present our novel framework for social media popularity prediction, the overall framework is shown in Figure 1. We extract features from many aspects, including the extraction of multimodal features, the extraction of time series features and the selection of multi-view important features.

2.1 MultiModal Features

Data from the Social Media Prediction Dataset (SMPD) can be roughly divided into three categories: images, texts and some digital or character data (i.g., location, time and user portrait data). For different types of data, we utilize different processing methods and extract different features.

2.1.1 Visual Feature Extraction. For the image data in the data set, we divide the data into 11 categories and 77 subcategories respectively according to the Category and Subcategory information of the posts it belongs to. Then, we utilize a pre-trained Shunted Transformer[18] model which has excellent ability to capture multi-scale objects to fine-tune the data of the two different hierarchies. Finally, we save the visual features of the two models' output after maximum average pooling before classification, and connect them in series, like:

$$f_{image} = [ST_{Category}, ST_{Subcategory}](image), \quad (1)$$

where ST refers to Shunted Transformer, $ST_{Category}$ refers to a trained model that classifies images by Category, and $ST_{Subcategory}$ refers to a trained model that classifies images by Subcategory.

2.1.2 Textual Feature Extraction. First, we clean the text data and delete non-alphabetic characters in the title of the text. Then, we mainly adopt two methods for textual feature extraction: 1) pre-trained model based, we use BERT-Large model[6], Fasttext[2], GloVe[17], CharNGram[10] and BPEmb[11] algorithms to directly extract textual features, where BERT and CharNGram use all texts together for textual feature extraction, while other algorithms use each input word for textual feature extraction; 2) statistics based, we utilize term frequency-inverse document frequency (TF-IDF)[19] and Latent Dirichlet Allocation (LDA)[1] algorithms to obtain the statistical features of text data. And we joint the text length, number of characters and number of words to enrich the statistical features. We calculate TF-IDF matrix and reduce dimension using TruncatedSVD[9] and train a topic model LDA with 20 topics to extract textual features according to [22]. Finally, we cascade these features together as the final textual feature.

$$f_{text} = [[BERT, FT, GloVe, CNG, BPE, TI, LDA](text), Others], \quad (2)$$

where FT refers to Fasttext, CNG refers to CharNGram, BPE refers to BPEmb, TI refers to TF-IDF, others refers to statistics such as text length.

2.1.3 Vision-Language Representation Feature Extraction. Because images and text are interactive and visual features and textual features are not aligned in semantic space, to make better use of visual and textual information, we use ALBEF[13] model pre-trained with triple contrastive learning[28] algorithm, which consists of an image encoder, a text encoder and a multimodal encoder to extract visual-language representation features. Among them, the image encoder is composed of 12 layers of ViT[7], while the text encoder and the multimodal encoder are composed of the first six layers and the last six layers of BERT-Base model, respectively. The output of the image encoder and the text encoder are fed into the multimodal encoder, and the textual features and the visual features are fused through the attention mechanism at each layer of the multi-modal encoder. Finally, the output of the multimodal encoder is our visual-language representation feature.

$$f_{vl} = ALBEF(image, text). \quad (3)$$

2.1.4 Numerical Feature Extraction. The numerical data includes two sources, one is provided by SMPD, and the other is additionally crawled from the Internet. For the data provided by SMPD, we convert the timestamp to year, month, day and hour and map it between 0 and 1. Meanwhile, we calculate the day of the week and save it as a categorical feature. Once processed, this part of the data is fed into CatBoost directly. For the other part, we first access user's website if the user provides a Pathalias, take Followers, Following, totalViews, totalTags, totalGeotagged, totalFaves and totalInGroup information from the website, and then constitute them as the numerical data of the second part.

2.2 Time Series Features

Timing is also very important for predicting the popularity of users' posts. Generally speaking, if one post is popular, then posts by the same author in the following period may also receive more attention, and have a high probability to be popular posts. Therefore,

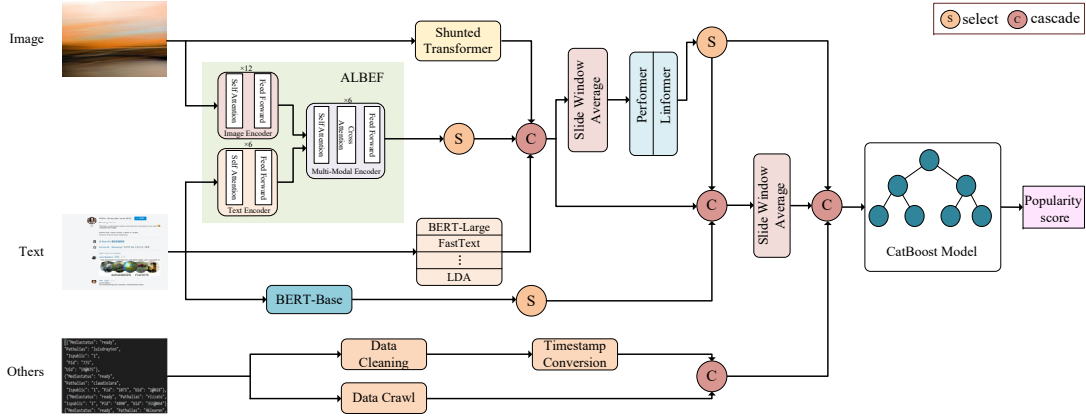


Figure 1: Overview of the proposed efficient multi-view multimodal data processing framework for social media popularity prediction.

it is of great significance to study how to make good use of timing characteristics. For SMPD, we adopt two methods to extract timing features, one is based on sliding window average, the other is based on Transformer[21].

2.2.1 Sliding Window Average. According to Wang’s method [22], the size of the window is set to 5, and the timing features of each post are the average value of the previous 4 posts and its own features of image and text that extracted by above methods. Afterward, the average feature $\bar{f}_{i,j}$ for the j_{th} post of i_{th} user can be defined as:

$$\bar{f}_{i,j} = \begin{cases} \frac{\sum_{k=0}^4 f_{i,j-k}}{5}, & j > 4 \\ \frac{\sum_{k=0}^{j-1} f_{i,j-k}}{j}, & j \leq 4 \end{cases}, \quad (4)$$

where f refers to the visual and textual features of the post.

As shown in Figure 1, we perform the algorithm twice in this paper. The first time is to obtain the input of the time series algorithm described below, and the second time is to obtain the sliding window average of all the features of images, text and time.

2.2.2 Transformer based Methods. In addition to using traditional methods, we also use Transformer based methods to extract timing features through attention mechanism, including Performer [5] and Linformer [23]. Attention mechanics are known for their high complexity, and Linformer and Performer can reduce that complexity to linearity while still performing better than native Transformer in sequence modeling.

Due to the large amount of SMPD, the training of deep learning model tends to train convergence for a long time. To save time and have better performance, we chose the two low-complexity Transformer models above for our timing feature extraction from different views.

After the features extracted by the sliding window in the first time are input, we set their sequence length to 10 and feed the features generated by sliding window average in the first time of the post in parallel with the previous 9 posts into model training to further extract the timing features. For partial inputs of length less than 10, we fill them with 0. We also use the popularity value as the training label, and apply the MSE loss function and the

SGD optimization algorithm for model training. However, we only have the popularity value for the training set, but not for the test set. Meanwhile, we can only train our model with the training set of SMPD, which will lead to the over-fitting of the model on the training set, i.e., timing features extracted from the training set have different distribution with the test set. However, the consistency of feature distribution between training set and test set is crucial for deep learning algorithm. Therefore, we divide the training set of SMPD into meta-training set and meta-validation set, calculate the loss function values of meta-training set and meta-validation set during the training process, and select the model weights with the closest two values and smaller loss function values as the results saved by our training. At the same time, in order to make the feature distribution of training set and test set more consistent, we set the ratio of meta-training set and meta-validation set to 1:9 instead of 9:1 in the conventional method, so that the saved model can make the feature distribution extracted from meta-training data consistent with that extracted from more meta-validation data.

We also train Performer and Linformer to categorize all the same features as above of the training set and test set based on the Sub-category information of the posts to extract timing features from another perspective, where the distribution of extracted features from the training set and test set was consistent. Noteworthy, we use features extracted from Performer and Linformer after feature importance selection, not directly. As shown in Figure 1, they cascade with other features before performing the second sliding window averaging, and then with the output features after performing it.

2.3 Multi-View Features

Using multi-view features jointly is very useful for the prediction task, but if we piece multi-view features together directly, the feature dimension may be very high and the model training will require too much memory that may exceed the total system memory. Thus, we selectively use some important features to save memory and improve the robustness of the proposed algorithm. Specifically, we first train several algorithms individually with CatBoost, then obtain the importance table of features generated by CatBoost after training, and finally select features with higher importance from the importance table. This way can greatly reduce the dimension

Feature	Average	
	SRC↑	MAE↓
image	0.2637	2.1711
text	0.5466	1.9734
VL	0.3571	2.0954
NF	0.0129	1.8556
P&L	0.5584	1.8338

Table 1: The contribution of each meta-part to prediction performance.

Feature	Average	
	SRC↑	MAE↓
image+text	0.5600	1.9363
image+text(slide)	0.5735	1.8556
image+text+VL	0.5587	1.9341
image+text+VL(slide)	0.5756	1.8482

Table 2: The difference between with and without sliding window average, and the difference between with and without VL. Slide refers to the previous feature performing a sliding window average.

of features. Then, we apply this strategy to ALBEF, Linformer, Performer and different layers' output of BERT-Base model, to make better use of multi-view features.

3 EXPERIMENTS

We use the Social Media Prediction Dataset (SMPD) to evaluate our proposed approach. SMPD contains 486K social multimedia posts from 70K users and various social media information including anonymous photo-sharing records, user profile, web image, text, time, location, category, etc. We use Spearman's Rho rank correlation (SRC) and Mean Absolute Error (MAE) as the evaluation metrics. All experimental results are shown in Table 1-4. In each table, "image" refers to visual features, "text" refers to textual features, "VL" refers to vision-language representation features, "NF" refers to Numerical features, "P&L" refers to the features extracted by Performer and Linformer after selection of importance.

First, we explore the influence of each meta-feature part on CatBoost model results. As shown in Table 1, features extracted by Performer and Linformer have the largest contribution for prediction, while numerical features have the smallest contribution. However, as shown in Table 3, we can observe that using numerical features can increase the value of SRC by 0.098 and decrease the value of MAE by 0.3516 (the second and fourth lines). This implies that numerical features are not very effective when used alone, but can be very effective if used with other features.

As shown in Table 2, we can observe that directly cascading vision-language representation features with other features will decrease the value of MAE, but lead to deterioration of SRC. However, VL cascaded features with sliding window average perform better than those without VL, whether SRC or MAE. We also evaluate the contribution of P&L. We can observe that P&L without participating in sliding window averaging outperforms P&L with participating in sliding window averaging. Meanwhile, adding P&L outperforms P&L without participating in sliding window averaging, regardless of whether or not it participates in sliding window averaging. Therefore, our final feature takes both of the approaches described above, as represented in the last line of Table 3.

Feature	Average	
	SRC↑	MAE↓
image+text+VL(slide)	0.5756	1.8482
image+text+VL(slide)+P&L	0.5938	1.7764
image+text+VL+P&L(slide)	0.5896	1.7841
image+text+VL(slide)+P&L+NF	0.6918	1.4248
image+text+VL+P&L(slide)+P&L+NF	0.6928	1.4249

Table 3: The contribution after adding the feature P&L and the contribution after adding NF, and also explores whether sliding window average should be performed for P&L.

Feature	Average					
	SRC(v)	MAE(v)	SRC(t)↑	MAE(t)↓	SRC(d)↓	MAE(d)↓
Performer(1)	0.5821	1.8727	0.5302	2.1328	0.0519	0.2601
Performer(2)	0.5195	2.0024	0.5173	2.1650	0.0022	0.1626
Performer(3)	0.8489	0.9533	0.4975	2.0389	0.3514	1.0856
Linformer(1)	0.5275	1.9619	0.5324	2.1368	0.0049	0.1749
Linformer(2)	0.5105	2.0112	0.5137	2.1773	0.0032	0.1661
Linformer(3)	0.7239	1.3964	0.4550	2.1504	0.2689	0.754

Table 4: The comparison of distribution consistency between our time feature extraction method and conventional method for training set and test set. (1) and (2) denote our first and second method, and (3) denote the conventional method. v and t represent the results on the validation set and test set respectively, and d denotes the difference between them.

Finally, we explore the effectiveness of our method to increase the distribution consistency of features of training set and test set when extracting time features with deep learning model. We split the training set and validation set in the ratio of 7:3 and then train the model with CatBoost. As shown in Table 4, we can observe that in the conventional method, the model has best performance on the validation set, but there is a big difference in the performance on the test set, where it is 0.3514 on SRC and 1.0856 on MAE. However, there is no significant difference between the results of our method in the validation set and test set. In terms of final performance, our method is better than the conventional method on SRC.

CONCLUSION

In this paper, we propose an efficient multi-view multimodal data processing framework for social media popularity prediction. Specifically, we design a feature extraction architecture that uses different algorithms to extract multi-view features, and use CatBoost to calculate the importance of different features and select the most important features for popularity prediction, which can reduce memory consumption, improve the representation of features to data, and improve the performance of the proposed approach. We also propose a Transformer and sliding window average based timing feature extraction method, which can reduce the inconsistent distribution between timing features extracted from training set and test set. Experimental results show that our approach achieves excellent performance for social media popularity prediction.

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